

30 Scenario-Based Questions

Topic: Variables + Operators Only

Student Task

For every question, students must write:

1. Required variables
2. Which operator will be used
3. Why that operator is used
4. Simple logic in words
5. Optional: small code

1. School Fee Calculator

A student's monthly fee is ₹1500. The student pays for 3 months together.

Think: Which variables are needed? Which operator calculates total fee?

2. Chocolate Sharing

A teacher has 40 chocolates and 8 students. Chocolates must be shared equally.

Think: Which operator finds chocolates per student?

3. Remaining Chocolates

A teacher has 43 chocolates and 8 students. After equal sharing, some chocolates remain.

Think: Which operator finds remaining chocolates?

4. Student Pass Check

A student scored 37 marks. Passing marks are 35.

Think: Which operator checks whether the student passed?

5. Age Eligibility

A student can join a coding workshop only if age is 12 or above.

Think: Which operator checks eligibility?

6. Shopping Total

A customer buys a bag for ₹700 and shoes for ₹1200.

Think: Which variables are needed? Which operator gives total bill?

7. Discount Calculation

A product price is ₹2000. Shop gives ₹300 discount.

Think: Which operator calculates final price?

8. Pocket Money Update

A student has ₹500. Parents give ₹200 more.

Think: Which assignment operator can update pocket money?

9. Pen Distribution

A box has 60 pens. Each student gets 5 pens.

Think: Which operator finds how many students can receive pens?

10. Even or Odd Roll Number

A roll number is 24.

Think: Which operator checks whether roll number is even?

11. Exam Result Message

Marks are 35. Passing marks are also 35.

Think: Should we use > or >=? Why?

12. Mobile Battery

Phone battery is 80%. After gaming, battery decreases by 25%.

Think: Which operator updates battery value?

13. Game Score

Player score is 100. Player collects a bonus coin worth 50 points.

Think: Which operator updates score?

14. Password Match

Correct password is python123. User enters a password.

Think: Which operator checks both passwords are same?

15. Wrong Password Check

Correct password is admin123. User enters admin.

Think: Which operator checks password is not correct?

16. Movie Ticket Cost

One movie ticket costs ₹180. Four friends buy tickets.

Think: Which operator calculates total ticket cost?

17. Fuel Cost

Bike uses 2 litres petrol. Petrol price is ₹105 per litre.

Think: Which operator calculates total petrol cost?

18. Temperature Check

Classroom temperature is 38°C. Normal limit is below 35°C.

Think: Which operator checks temperature is high?

19. Height Comparison

Student A height is 150 cm. Student B height is 145 cm.

Think: Which operator checks who is taller?

20. Same Marks Check

Two students both scored 88 marks.

Think: Which operator checks marks are equal?

21. Attendance Eligibility

A student has 74% attendance. Exam eligibility is 75% or above.

Think: Which operator checks eligibility?

22. Book Price After Offer

A book costs ₹500. Offer reduces price by ₹50.

Think: Which operator calculates new book price?

23. Monthly Savings

A student saves ₹100 every week. Find savings after 4 weeks.

Think: Which operator calculates total savings?

24. Water Bottles

There are 25 water bottles. Each table gets 4 bottles.

Think: Which operator finds complete tables served? Which finds leftover bottles?

25. Classroom Chairs

A class has 36 students and 30 chairs.

Think: Which operator checks if chairs are enough?

26. Simple Salary Update

Employee salary is ₹15000. Salary increases by ₹2000.

Think: Which assignment operator updates salary?

27. Online Cart Quantity

Cart has 2 notebooks. Student adds 3 more notebooks.

Think: Which operator updates quantity?

28. Comparing Product Prices

Pen price is ₹10. Pencil price is ₹5.

Think: Which operator checks if pen is costlier?

29. Marks Percentage

A student scored 420 out of 500.

Think: Which operators calculate percentage?

30. Final Bill With GST

A product costs ₹1000. GST is 18%.

Think: Which operators calculate GST amount and final bill?

ADVANCE Level 1

1.

A food delivery app gives free delivery only when the order amount is above ₹499 and the user has premium membership.

2.

An ATM allows withdrawal only if the account balance is greater than the withdrawal amount.

3.

A school gives scholarships to students scoring more than 85 marks or students who won a national competition.

4.

A movie theatre charges double ticket price during weekends.

5.

An online exam portal allows login only when both username and password are correct.

6.

A shopping app adds 18% GST after applying discount on the product price.

7.

A game reduces player health by 20 points after every enemy attack.

8.

A hospital allows emergency treatment immediately if the patient age is below 10 or above 60.

9.

A parking system allows entry only if parking slots are available.

10.

A student can participate in sports only if attendance is 75% or above.

11.

An e-commerce app gives extra cashback when the purchase amount crosses ₹10,000.

12.

A railway booking system checks whether seats are available before confirming booking.

13.

A restaurant bill is divided equally among 5 friends. Remaining amount, if any, is paid by one friend.

14.

A smart door unlocks only if face recognition and fingerprint verification both succeed.

15.

A YouTube creator earns bonus income when subscriber count reaches 100000.

16.

An electricity bill increases by ₹8 per extra unit after crossing 100 units.

17.

A cricket player is selected only if runs are above 500 and fitness score is above 80.

18.

A bank gives loan approval only when salary is above ₹30,000 and credit score is above 700.

19.

A teacher checks whether a roll number belongs to an even bench or odd bench.

20.

A mobile app allows password reset only if entered OTP matches system OTP.

21.

A gym gives premium membership discounts to users below age 25 or above age 55.

22.

An online shopping cart updates total amount whenever a new item is added.

23.

A fuel station calculates total petrol cost based on litres entered by customer.

24.

A laptop battery warning appears when battery percentage falls below 15%.

25.

A company calculates yearly bonus by multiplying monthly salary with performance multiplier.